



Largest prime number

The world has a new largest ever known prime number and it is nearly 25 million digits long. <https://digit.in/jan19-77>



'Super Earth'

Astronomers have found a new shimmering planet or 'Super Earth' in the constellation called Cassiopeia. <https://digit.in/jan19-78>



Seen here are a number of fairy circles in the Namib desert. If you want to check out the exact location on Google Maps, use these co-ordinates: -24.13, 15.69

the total number of circles, which are mostly in a state of stasis. This led to a belief among early researchers that fairy circles are permanent features of the desert. This has now been disproved, and the life cycles of fairy circles are fairly well understood.

The Namibian fairy circles are just one type of regular patterns in vegetation seen around the world. There are Mima mounds in North America and the heuweltjies in South Africa, both of which have not been totally explained by science.

SPATIAL SELF ORGANISATION

There have been many proposed explanations for the fairy circles, many of which have been disproved by experiments. Some proposed explanations include termite mounds that accumulate water in rings, underground emissions of poisonous gases, or toxins left behind by plants that are dead. The murundus in Brazil, another natural pattern, are engineered by termites. One study published in 2013 found termite activity in 80 percent of fairy circles surveyed, and 100 percent of new circles. However, there have been fairy circles found in places where termites did not exist at all, and the mere existence of termites does not prove that they cause the fairy circles in the first place. The fairy circles have persisted even when underground rubber sheets have been introduced beneath the soil. Researchers were able

to take samples of the barren ground to laboratories, and grow plants in them, showing that the soil itself was not toxic. Analysis of the soil showed no toxins, had the normal amount of nutrients, and was not radioactive.

Another hypothesis is that the patterns of the fairy circle emerge naturally because of competition among the plants for a limited vegetation resources. In arid landscapes, where there is limited water available, vegetation grows in patches or bands all around the world. According to the theory, the grasses start expanding outwards in search of water. The individual plants also compete with one another below the ground. This competition results in the elimination of neighbours, forming the ring like structures. All the individual plants in a ring compete to derive nutrition from a common pool of water in the middle of the circles. As the circles ma-

ture, the plants at the periphery of the ring begin drawing the water that has seeped into the ground. These plants create underground reservoirs of water, that further encourage more fit plants to grow. The older dead plants, also deposit nutrients around the periphery of the circles. This creates a positive feedback loop over multiple generations, of an underground resource pool of water and nutrients. The fairy circles can grow outward through this positive feedback loop. The sizes and spacings of the rings can be explained by inter-ring competition. This kind of spatial self organisation by plants has been supported by mathematical modelling.

In 2017, a study published in *Nature* proposed a one-two punch as an explanation for the formation of fairy circles. According to the research, the competition among the plants over limited resources or activity by termites could not explain the regular vegetation patterns in isolation. The theory proposes that the activity of both termites and the positive feedback loops together create very resilient circles, that can survive periods of drought.

In 2018, a group of scientists found fairy circles in the remote outback of Australia. The fairy circles appeared in an arid landscape, and provided further support for the theory of self organisation and positive feedback loops. The finding further suggests that there may be even more fairy circles waiting to be discovered in remote deserts all around the world.

Fairy circles are an important part of the local ecosystem. They support the rummaging fauna, as well as the livestock of the indigenous peoples. The locals at times fence off their animals in temporary fields of the fairy circles, to keep the animals safe from predators. The novelty of the circles are used towards conservation efforts. Visitors can adopt individual circles (for about 5000), by donating to the Namib Rand Nature Reserve, a private local foundation. A numbered plaque is placed next to the circle in the name of the sponsor. The fairy circles of Namibia remain one of the unsolved mysteries of science. 



A Namibian Oryx sitting within a fairy circle

Credit: NamibRand Nature Reserve